1. K-Nearest Neighbors (KNN)

1. A KNN model compares two customers using age and weekly usage. Which equation gives the Euclidean distance between customers A = (a_1, u_1) and B = (a_2, u_2)?
A. $d(A,B) = \sqrt{[(a_1 - a_2)^2 + (u_1 - u_2)^2]}$
B. $d(A,B) = (a_1 + a_2) / (u_1 + u_2)$
C. $d(A,B) = a_1u_1 + a_2u_2$
D. $d(A,B) = |a_1 - a_2| \times |u_1 - u_2|$

2. A new customer's distances to three training customers are 2, 5, and 8. Their churn labels are 1, 0, and 0, respectively. For K = 1, what does the KNN voting equation use?
A. The label of the customer at distance 2
B. The majority label among all three customers
C. The average of the three distances
D. The label of the customer at distance 8

3. For KNN regression with K = 3 and uniform weights, the three nearest houses have prices of ₹40 lakh, ₹50 lakh, and ₹60 lakh. Which expression gives the predicted price?
A. (40 + 50 + 60) / 3 lakh
B. (40 × 50 × 60) / 3 lakh
C. 40 + 50 + 60 lakh
D. max(40, 50, 60) − min(40, 50, 60) lakh

4. A KNN model uses Manhattan distance for two features. Customer A has values (4, 9), and customer B has values (7, 5). Which expression calculates their distance?
A. $|4 - 7| + |9 - 5|$
B. $\sqrt{[(4 - 7)^2 + (9 - 5)^2]}$
C. $(4 + 7) \times (9 + 5)$
D. $(4 - 7) \times (9 - 5)$

5. A dataset contains customer age ranging from 18 to 70 and monthly fee ranging from ₹200 to ₹1,500. KNN uses a distance equation containing both features. Why might scaling be necessary?
A. The larger numerical range of monthly fee could dominate the distance
B. Scaling changes the churn labels from 0 and 1 to customer IDs
C. Euclidean distance cannot be calculated with two features
D. Scaling makes every customer's distance equal

2. Naive Bayes

1. A model classifies an email as spam (S) using its words (W). Which expression represents the probability the model wants to estimate?
A. P(S | W)
B. P(W | S) only
C. P(S) × P(W) only
D. P(S + W)

2. Bayes' theorem can be written as P(S | W) = P(W | S)P(S) / P(W). In this equation, what does P(S) represent?
A. The prior probability of spam
B. The probability of the words given spam
C. The distance to the nearest spam email
D. The model's classification accuracy

3. In a training set of 200 emails, 50 are spam. If the model estimates the spam prior from this dataset, which expression does it use?
A. P(spam) = 50 / 200
B. P(spam) = 200 / 50
C. P(spam) = 50 × 200
D. P(spam) = 200 − 50

4. A simplified Naive Bayes calculation uses $P(x_1, x_2 \mid \text{class}) \approx P(x_1 \mid \text{class}) \times P(x_2 \mid \text{class})$. What assumption allows this multiplication?
A. The features are conditionally independent given the class
B. The two features always have identical values
C. Both classes have equal numbers of samples
D. The features are continuous rather than categorical

5. A word never appeared in spam emails during training. The model uses Laplace smoothing: P(word | spam) = (count(word in spam) + α) / (total spam word count + αV), where V is vocabulary size. Why is α added?
A. To avoid assigning zero probability solely because the word was unseen
B. To increase the number of spam emails in the dataset
C. To calculate Euclidean distance
D. To set the classification threshold to α

3. Decision Trees

1. A tree node contains 8 customers who stayed and 2 who canceled. Its Gini impurity is calculated as Gini = 1 − (8/10)² − (2/10)². What does this equation measure?
A. How mixed the two classes are at the node
B. The distance between the ten customers
C. The number of branches in the whole tree
D. The probability that every customer will cancel

2. A decision tree considers splitting a node using weekly_usage_hours <= 3. What is the purpose of comparing impurity before and after the split?
A. To assess whether the split creates more useful class separation
B. To calculate the average age of all customers
C. To select the nearest customer
D. To determine the number of trees in a forest

3. Entropy for two classes can be written as $H = -p_0 \log_2(p_0) - p_1 \log_2(p_1)$. If a leaf contains only customers who stayed, which values describe its class proportions?
A. $p_0 = 1$ and $p_1 = 0$
B. $p_0 = 0.5$ and $p_1 = 0.5$
C. $p_0 = 1$ and $p_1 = 1$
D. $p_0 = 0$ and $p_1 = 0$

4. A split produces two child nodes containing 30 and 10 customers. Which expression calculates their weighted average impurity?
A. (30/40) × impurity(left) + (10/40) × impurity(right)
B. impurity(left) + impurity(right) + 40
C. 30 × 10 × impurity(left)
D. [impurity(left) − impurity(right)] / 40

5. A customer reaches a tree leaf containing 12 customers who canceled and 8 who stayed. Which expression could be used as that leaf's estimated churn probability?
A. 12 / (12 + 8)
B. 8 / 12
C. (12 + 8) / 12
D. 12 × 8

4. Random Forest

1. Five trees predict churn labels [1, 1, 0, 1, 0] for a customer. A random forest uses majority voting. Which expression describes the final class?
A. mode(1, 1, 0, 1, 0)
B. max(1, 1, 0, 1, 0) − 1
C. 1 × 1 × 0 × 1 × 0
D. The label predicted by the first tree only

2. Four trees in a random forest regressor predict customer spending of ₹800, ₹1,000, ₹900, and ₹1,100. Which expression gives the forest's usual regression prediction?
A. (800 + 1,000 + 900 + 1,100) / 4
B. 800 × 1,000 × 900 × 1,100
C. max(800, 1,000, 900, 1,100)
D. 1,100 − 800

3. A forest contains 100 trees. For a particular customer, 72 trees vote for churn. Which simple fraction represents the proportion of trees voting for churn?
A. 72 / 100
B. 100 / 72
C. 100 − 72
D. 72 × 100

4. A forest is trained using bootstrap samples. Which equation-like description matches sampling *with replacement*?
A. A training customer can appear more than once in one tree's sample
B. Each training customer must appear exactly once in every tree's sample
C. Every tree trains only on the test set
D. Training customers are selected according to their predicted labels

5. A tree's out-of-bag prediction is evaluated using training customers excluded from *that tree's* bootstrap sample. Which expression identifies an out-of-bag customer for that tree?
A. Customer ∈ training data AND customer ∉ that tree's bootstrap sample
B. Customer ∈ final test set only
C. Customer ∈ that tree's bootstrap

5. Logistic Regression

1. A logistic regression model computes $z = b_0 + b_1x_1 + b_2x_2$ and then p = 1 / (1 + e^(−z)). What does p represent in a binary churn model?
A. The estimated probability of the positive class, such as churn
B. The number of trees required
C. The distance to the nearest customer
D. The number of input features

2. A model uses a threshold of 0.5 and estimates P(churn = 1 | X) = 0.73. Which rule assigns its class label?
A. Predict churn if p ≥ 0.5
B. Predict churn only if p = 1
C. Predict churn if p < 0.5
D. Predict churn only if every feature is positive

3. The model's linear score is z = −2 + 0.5 × support_tickets. For a customer with 4 support tickets, which expression gives z?
A. −2 + 0.5 × 4
B. −2 / (0.5 × 4)
C. −2 × 0.5 × 4

D. 4 / (−2 + 0.5)

4. The odds of churn are defined as p / (1 − p). If p = 0.75, which expression gives the odds?
A. 0.75 / 0.25
B. 0.25 / 0.75
C. 0.75 + 0.25
D. 0.75 × 0.25

5. Logistic regression can be written as $\log[p / (1 - p)] = b_0 + b_1x_1 + \ldots + b_nx_n$. A fitted coefficient b_1 is positive. Holding the other features fixed, what does increasing x_1 do?
A. It increases the predicted log-odds of the positive class
B. It always decreases the predicted log-odds
C. It increases the number of training samples
D. It changes the model into a decision tree